

Editor-in-Chief
Image and Vision Computing

Dear Editor-in-Chief,

We are pleased to submit the manuscript entitled “GRA-CNN: Semantic-Aware Structured Channel Pruning via Gray Relational Analysis” for consideration as an original research article in *Image and Vision Computing*.

The manuscript addresses a practical computer-vision problem: how to prune convolutional recognition models in a way that preserves decision-relevant visual information under aggressive compression. Existing structured pruning criteria are typically driven by local structural cues such as filter magnitude, geometry, or rank. Our work introduces GRA-CNN, a two-stage pruning framework that combines a structural anchor with a quality-gated semantic refinement based on Gray Relational Analysis. The central idea is to evaluate whether a channel is redundant in a class-aware manner, rather than relying exclusively on local structural weakness.

We believe the work fits *Image and Vision Computing* for three reasons. First, the contribution is squarely located in visual model design and efficient visual recognition. Second, the method provides an algorithmic innovation that is specific to vision backbones and their activation behavior. Third, the experimental study is deliberately rigorous: it includes multiple architectures, multiple compression regimes, multi-seed evaluation, matched-budget comparisons, a transfer study on Tiny-ImageNet, and supporting mechanism-level analysis.

Importantly, we position the contribution carefully. The paper does not claim that semantic scoring is universally superior for every architecture and operating point. Instead, the evidence shows that GRA-CNN is most useful as a robustness-oriented refinement in high-compression residual CNN settings, while remaining competitive and transparent in more architecture-dependent cases. We believe this balanced reporting style strengthens the paper’s relevance for the IVC readership.

This manuscript has not been published previously and is not under consideration for publication elsewhere. All authors have approved the manuscript and agree with its submission to *Image and Vision Computing*.

Thank you for your time and consideration.

Sincerely,

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